

Capstone Project: Full Alert-to-Response Cycle

Title

Full SOC Alert-to-Response Cycle Simulation

Course: Cybersecurity Operations

Week: Week 2

Prepared By: SOC Analyst

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1. Executive Summary

A simulated cyberattack was conducted against a vulnerable server running Metasploitable2. The attack exploited the VSFTPD backdoor vulnerability using Metasploit. Security monitoring tools detected suspicious activity, allowing the SOC team to investigate, contain, eradicate, and recover from the incident.

The exercise demonstrated the complete incident response lifecycle from attack detection to final reporting.

The incident was successfully contained with no significant business impact.

2. Objectives

The objectives of this capstone project were:

- Simulate a realistic cyberattack.
- Detect malicious activity.
- Perform alert triage.
- Conduct incident response.
- Preserve evidence.
- Document findings.
- Produce management reports.

3. Environment Setup

Attacker Machine

Item	Value
Operating System	Kali Linux
IP Address	192.168.1.100
Tool Used	Metasploit

Target Machine

Item	Value
Operating System	Metasploitable2
IP Address	192.168.1.10
Vulnerability	VSFTPD Backdoor

Monitoring Tools

Tool	Purpose
Wazuh	Security Monitoring
VirusTotal	IOC Validation
CrowdSec	Threat Blocking
TheHive	Incident Management

4. Attack Simulation

Attack Description

The attacker attempted to exploit the VSFTPD 2.3.4 backdoor vulnerability.

Exploit Used

use exploit/unix/ftp/vsftpd_234_backdoor

Target Configuration

set RHOSTS 192.168.1.10

Launch Attack

run

Result

The exploit successfully established unauthorized access to the target machine.

MITRE ATT&CK Mapping

Technique	ID
Exploit Public Facing Application	T1190
Command Execution	T1059
Persistence	T1547

5. Detection Phase

Security Alert Generated

Wazuh generated an alert after detecting suspicious FTP activity and unauthorized access attempts.

Alert Information

Timestamp	Source IP	Destination IP	Alert
11:00 AM	192.168.1.100	192.168.1.10	VSFTPD Exploit Attempt

Severity

Critical

Initial Findings

- Multiple suspicious FTP connections.
- Unauthorized shell access detected.
- Known vulnerable service identified.

6. Alert Triage

Triage Process

Step 1

Verify source and destination.

Source:

192.168.1.100

Target:

192.168.1.10

Step 2

Review logs.

Finding:

FTP Backdoor Exploitation Attempt

Step 3

Validate Indicators.

IOC validation performed using threat intelligence resources.

Step 4

Determine Severity.

Impact:

High

Likelihood:

High

Priority:

Critical

Final Classification

True Positive

7. Incident Response

Identification

SOC analyst confirmed active exploitation.

Containment

Actions performed:

- Isolated vulnerable server.
- Blocked attacker IP.
- Disabled affected service.

Containment Evidence

Blocked IP:

192.168.1.100

Eradication

Actions:

- Removed malicious sessions.
- Patched vulnerable software.
- Updated security controls.

Recovery

Actions:

- Restored affected services.
- Verified server integrity.
- Re-enabled network connectivity.

Lessons Learned

Actions:

- Improve patch management.
- Enhance vulnerability scanning.
- Increase monitoring coverage.

8. Evidence Collection

Evidence Inventory

Evidence ID	Description
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EV-001	Wazuh Alert Logs
EV-002	Network Traffic Logs
EV-003	Firewall Logs
EV-004	System Audit Logs
EV-005	Incident Response Notes

Hash Verification

Evidence	SHA256
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Alert Log	SHA256-ABC123
Firewall Log	SHA256-XYZ456

9. Incident Timeline

Time	Activity
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11:00 AM	Exploit launched
11:02 AM	Wazuh generated alert
11:05 AM	Analyst assigned
11:10 AM	Alert triage completed
11:15 AM	Incident confirmed
11:20 AM	Server isolated
11:45 AM	Threat removed
12:15 PM	Recovery initiated
01:00 PM	Incident closed

10. Root Cause Analysis

Root Cause

The target server contained an outdated and vulnerable version of VSFTPD susceptible to public exploitation.

Contributing Factors

- Missing patches.
- Weak vulnerability management.
- Legacy services enabled.

11. Recommendations

Immediate Actions

- Patch vulnerable services.
- Remove unnecessary services.
- Block malicious IPs.

Short-Term Improvements

- Conduct vulnerability assessments.
- Improve alert tuning.
- Enable MFA where applicable.

Long-Term Improvements

- Implement Zero Trust Architecture.
- Deploy EDR solutions.
- Perform regular penetration testing.

12. Stakeholder Briefing (Non-Technical)

Management Summary

A cybersecurity incident involving an attempted exploitation of a vulnerable server was detected and successfully contained by the Security Operations Center. The attack targeted a known software vulnerability and resulted in unauthorized access attempts. Security monitoring systems identified the activity immediately, allowing analysts to isolate the affected server and prevent further compromise. No sensitive data was lost, and business operations remained unaffected. The vulnerable software was patched, monitoring controls were improved, and additional security recommendations were provided to reduce future risks. The incident has been fully resolved and normal operations have resumed.

13. Conclusion

This capstone project demonstrated the complete SOC workflow from attack simulation through detection, triage, containment, eradication, recovery, and reporting. The exercise highlighted the importance of continuous monitoring, rapid response, proper evidence handling, and thorough documentation. Through effective incident response procedures, the threat was successfully mitigated and valuable lessons were learned to strengthen organizational security.